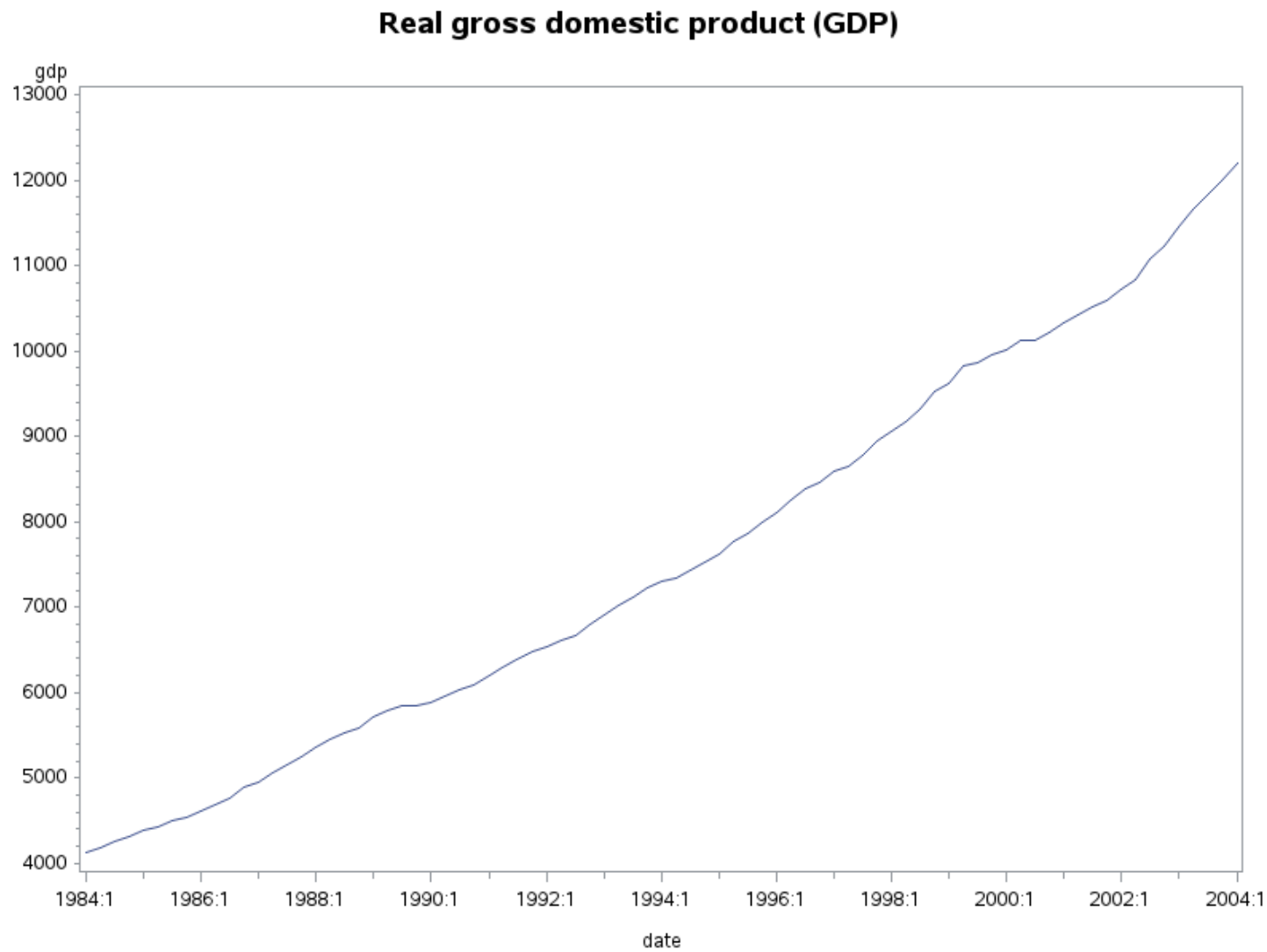
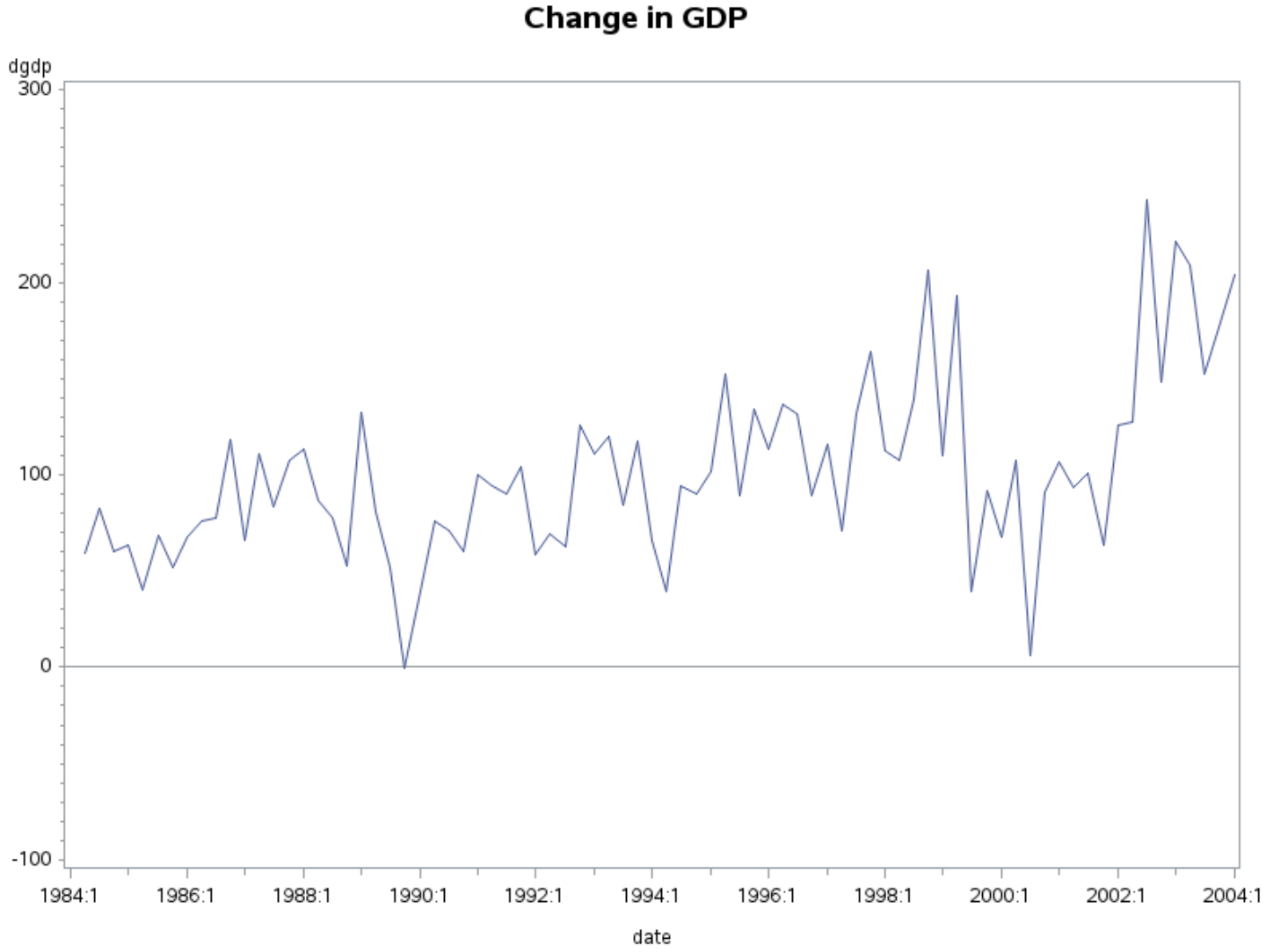
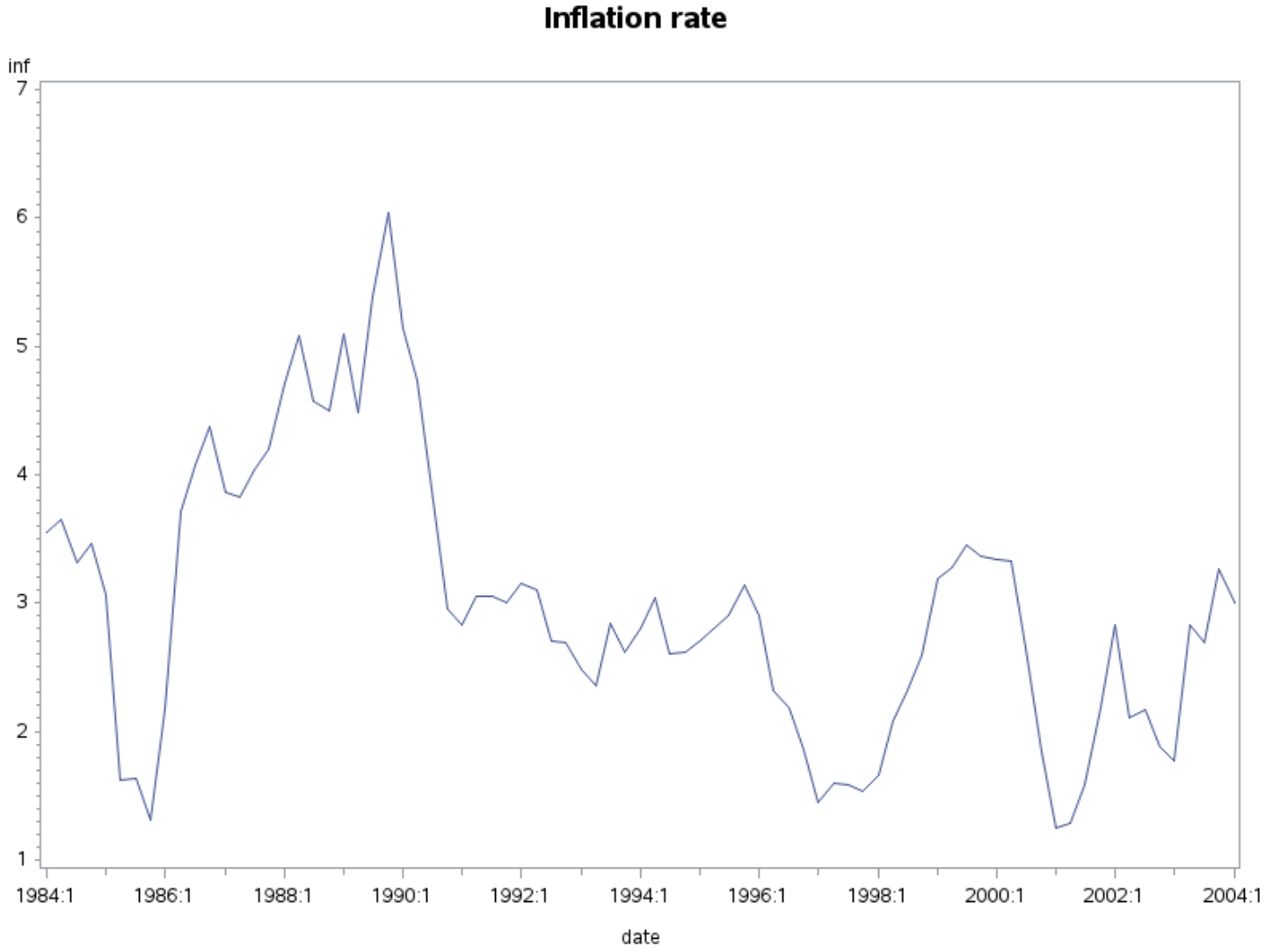
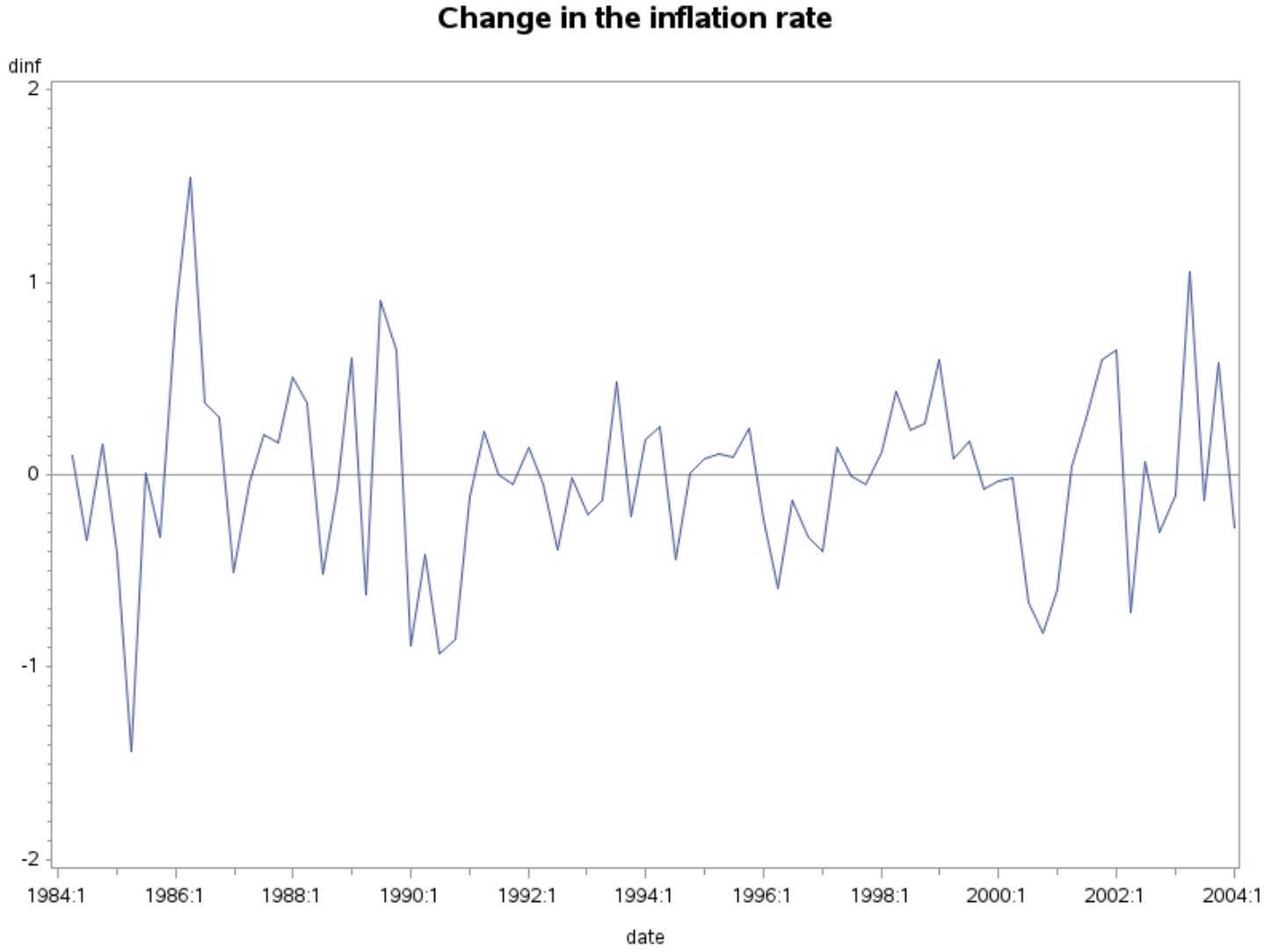
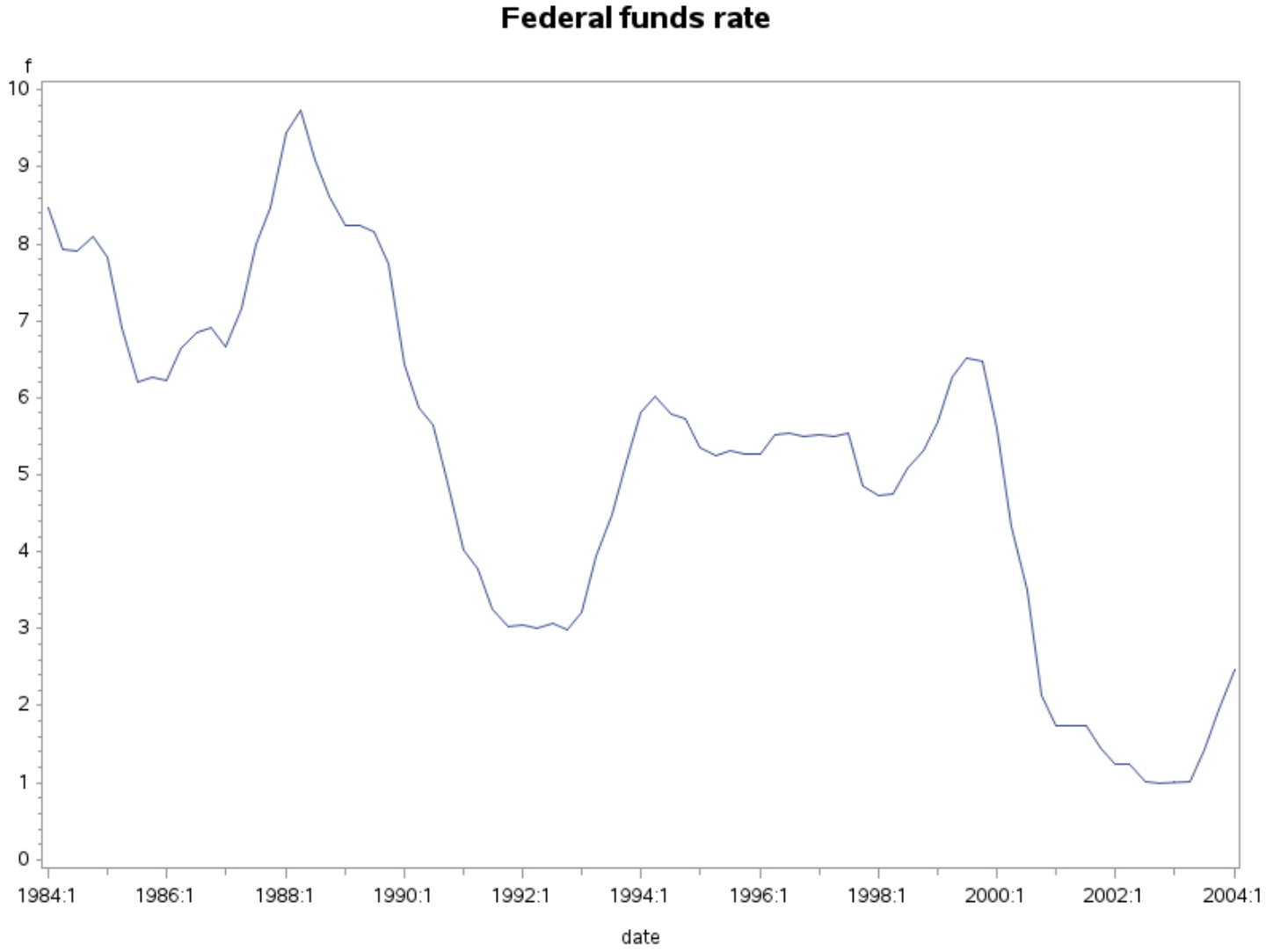
Obs	gdp	inf	f	b	dgdp	dinf	df	db	date	year	qtr
1	4119.5	3.54862	8.47667	10.6767	.	.	.	.	1984:1	1984	1
2	4178.4	3.64569	7.92333	9.7633	58.9	0.09707	-0.55334	-0.91337	1984:2	1984	2
3	4261.3	3.30914	7.90000	9.2867	82.9	-0.33655	-0.02333	-0.47666	1984:3	1984	3
4	4321.8	3.46945	8.10333	8.8433	60.5	0.16031	0.20333	-0.44334	1984:4	1984	4
5	4385.6	3.06095	7.82667	7.9367	63.8	-0.40850	-0.27666	-0.90666	1985:1	1985	1

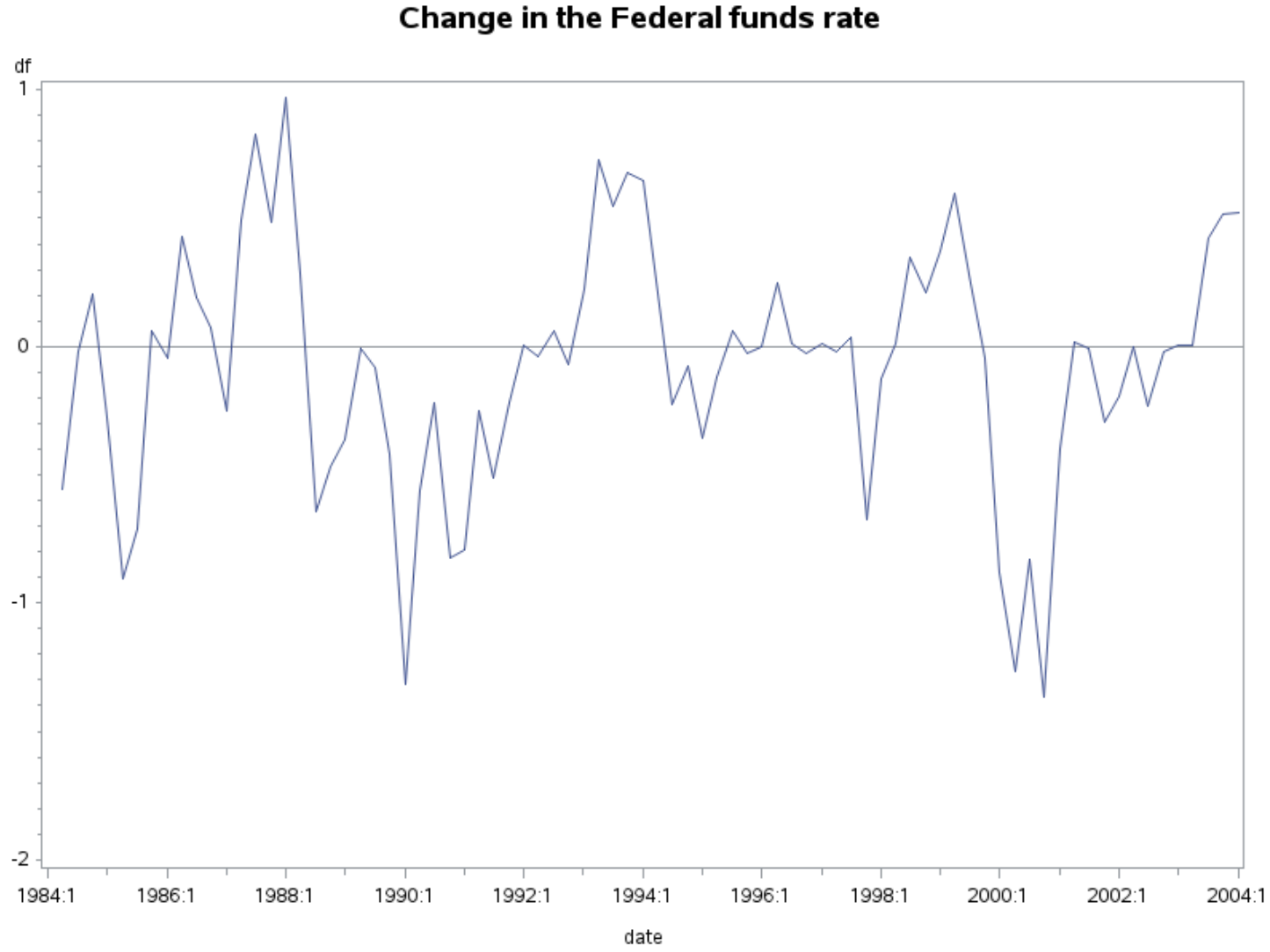


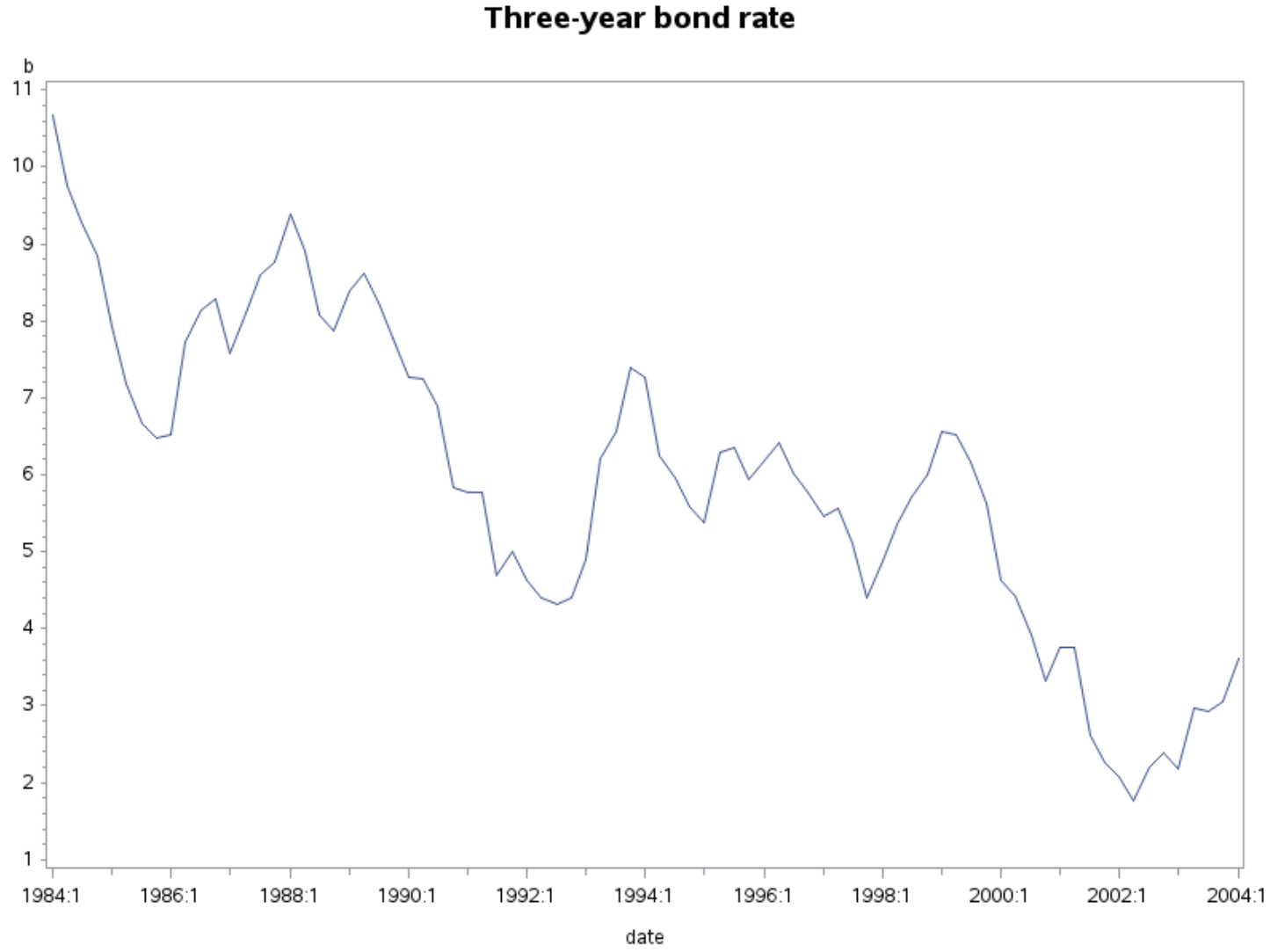

Change in GDP
dgdp
300
200
100
0
-100
1984:1
1986:1
1988:1
1990:1
1992:1
1994:1
1996:1
1998:1
2000:1
2002:1
2004:1
date

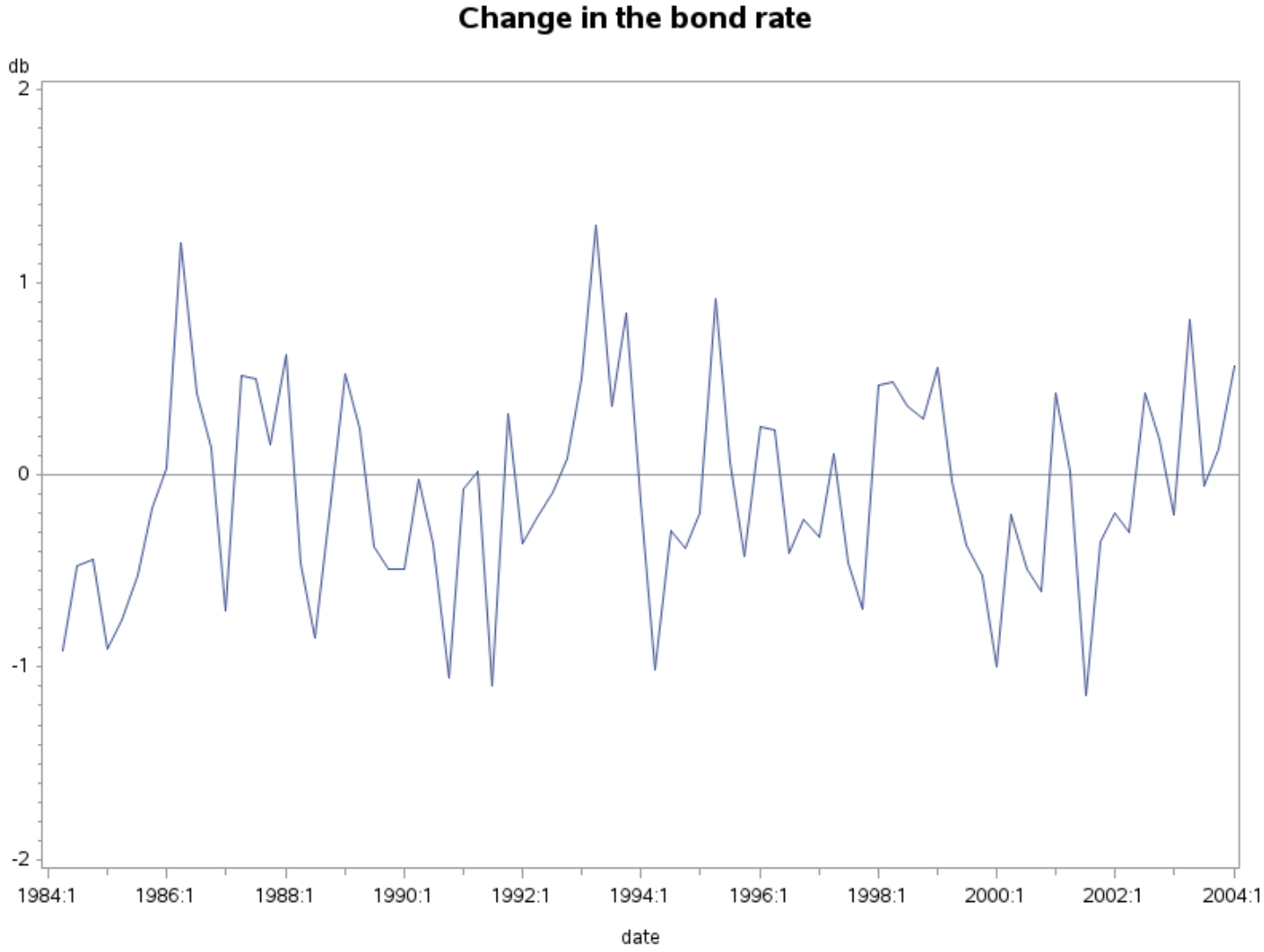

Inflation rate
inf
7
6
5
4
3
2
1
1984:1
1986:1
1988:1
1990:1
1992:1
1994:1
1996:1
1998:1
2000:1
2002:1
2004:1
date


Change in the inflation rate
dinf
2
1
0
-1
-2
1984:1
1986:1
1988:1
1990:1
1992:1
1994:1
1996:1
1998:1
2000:1
2002:1
2004:1
date


Federal funds rate
f
10
9
8
7
6
5
4
3
2
1
0
1984:1
1986:1
1988:1
1990:1
1992:1
1994:1
1996:1
1998:1
2000:1
2002:1
2004:1
date


Change in the Federal funds rate
df
1
0
-1
-2
1984:1
1986:1
1988:1
1990:1
1992:1
1994:1
1996:1
1998:1
2000:1
2002:1
2004:1
date


Three-year bond rate
b
11
10
9
8
7
6
5
4
3
2
1
1984:1
1986:1
1988:1
1990:1
1992:1
1994:1
1996:1
1998:1
2000:1
2002:1
2004:1
date



summary statistics 1984q2 to 1996q4

The MEANS Procedure

Variable	N	Mean	Std Dev	Minimum	Maximum
gdp	51	6146.35	1247.62	4178.40	8471.20
inf	51	3.3444352	1.0523371	1.3083100	6.0375670
f	51	6.0671894	1.8451824	2.9900000	9.7266700
b	51	6.8984971	1.4364833	4.3200000	9.7633300
dgdp	51	85.3274510	30.3102562	-0.6000000	152.5000000
dinf	51	-0.0332198	0.5059833	-1.4346830	1.5418620
df	51	-0.0582353	0.4664334	-1.3166600	0.9733300
db	51	-0.0960137	0.5595135	-1.0966700	1.2966700

summary statistics 1997q1 to 2009q4

The MEANS Procedure

Variable	N	Mean	Std Dev	Minimum	Maximum
gdp	29	10232.44	1021.28	8586.70	12198.80
inf	29	2.3475549	0.7266561	1.2437890	3.4482480
f	29	3.4768961	2.0398114	0.9966670	6.5200000
b	29	4.1117241	1.4951050	1.7733300	6.5566700
dgdp	29	128.5379310	56.5762975	6.2000000	242.8000000
dinf	29	0.0394029	0.4377375	-0.8200160	1.0574520
df	29	-0.1047128	0.4948690	-1.3633400	0.5966600
db	29	-0.0747128	0.4881009	-1.1500000	0.8033400

augmented Dickey-Fuller test for federal funds rate

The AUTOREG Procedure

Dependent Variable	df

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augmented Dickey-Fuller test for federal funds rate

The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	9.54349001	DFE	76
MSE	0.12557	Root MSE	0.35436
SBC	70.327132	AIC	63.2187885
MAE	0.25894806	AICC	63.5387885
MAPE	336.919146	HQC	66.0666084
Durbin-Watson	1.8551	Total R-Square	0.4560

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
Intercept	1	0.1779	0.1008	1.77	0.0815
f1	1	-0.0371	0.0177	-2.09	0.0399
df1	1	0.6725	0.0854	7.88	<.0001

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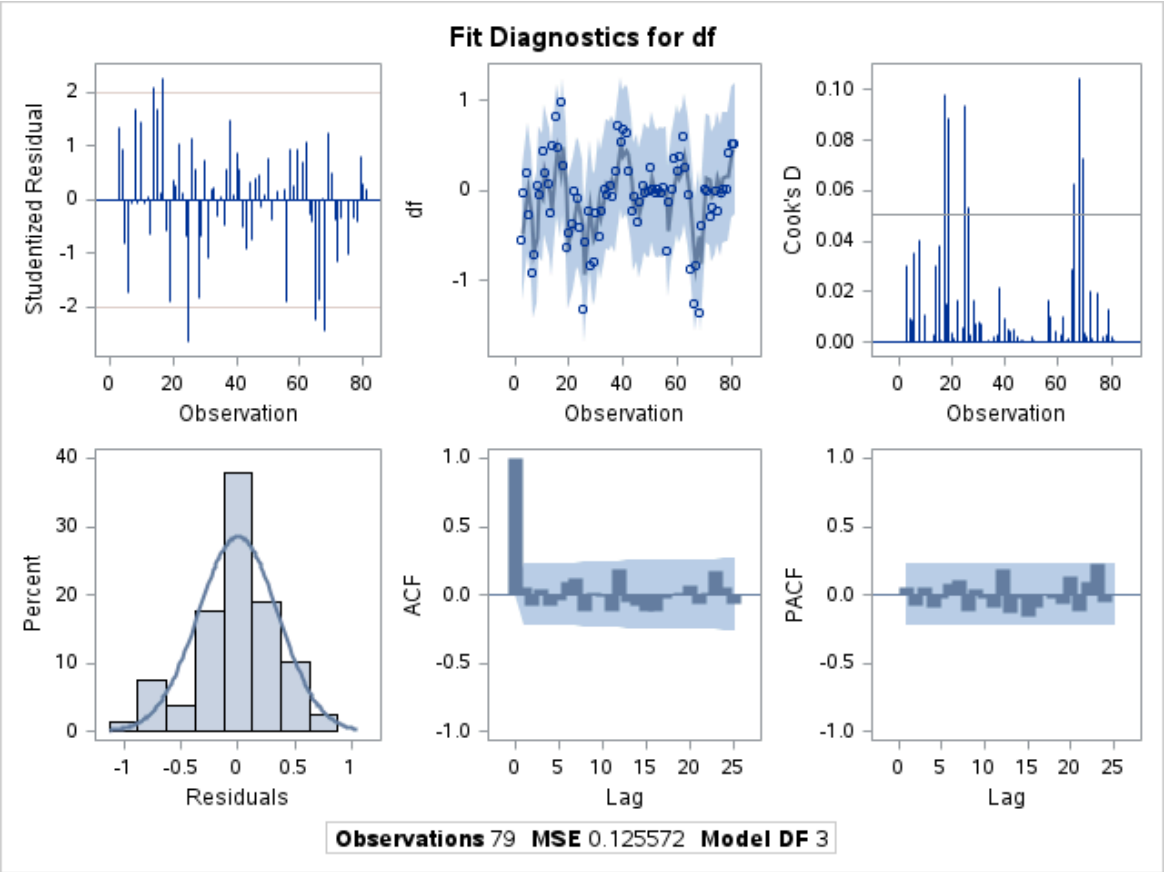
augmented Dickey-Fuller test for federal funds rate

The AUTOREG Procedure



Dickey-Fuller test for federal funds rate

Obs	_TYPE_	_STATUS_	_DEPVAR_	_NAME_	_MSE_	Intercept	AR_V	DLAG_V	AR_V1	_NOBS_	_TAU_	_TREND_	_DLAG_	_PVALUE_
1	PARM	0 Converged	AR_V		0.12557	0.17786	-1	-0.037067	0.67248	79	-2.09030	1	1	0.24892
2	COV	0 Converged	AR_V	Intercept	0.12557	0.01015	.	-0.001636	0.00141	79	-2.09030	1	1	0.24892
3	COV	0 Converged	AR_V	DLAG_V	0.12557	-0.00164	.	0.000314	-0.00016	79	-2.09030	1	1	0.24892
4	COV	0 Converged	AR_V	AR_V1	0.12557	0.00141	.	-0.000156	0.00729	79	-2.09030	1	1	0.24892

augmented Dickey-Fuller test for 3 year bond rate

The AUTOREG Procedure

Dependent Variable	db

augmented Dickey-Fuller test for 3 year bond rate

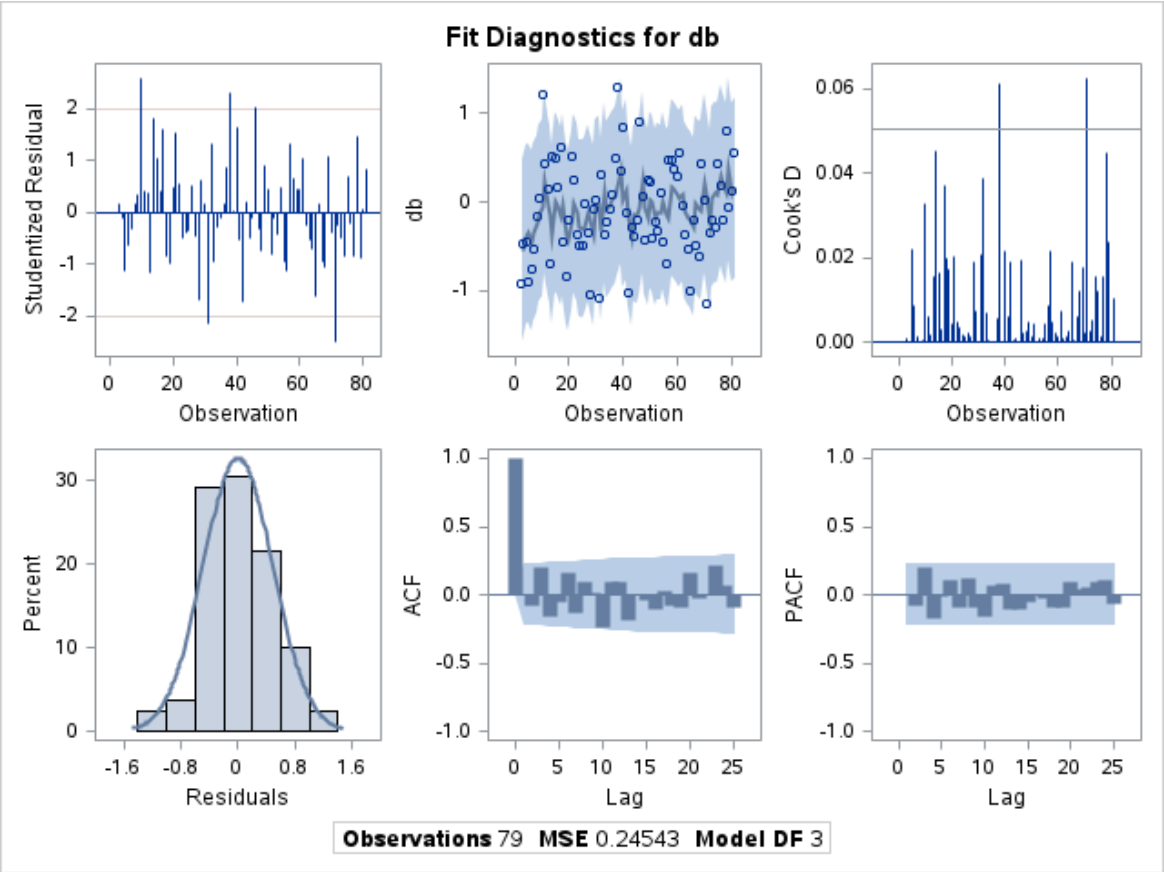
The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	18.6526528	DFE	76
MSE	0.24543	Root MSE	0.49541
SBC	123.267333	AIC	116.15899
MAE	0.38633299	AICC	116.47899
MAPE	141.913204	HQC	119.006809
Durbin-Watson	1.9751	Total R-Square	0.1382

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
Intercept	1	0.2852	0.1781	1.60	0.1134
b1	1	-0.0562	0.0284	-1.98	0.0517
db1	1	0.3151	0.1060	2.97	0.0040

augmented Dickey-Fuller test for 3 year bond rate

The AUTOREG Procedure



Dickey-Fuller test for 3 year bond rate

Obs	_TYPE_	_STATUS_	_DEPVAR_	_NAME_	_MSE_	Intercept	AR_V	DLAG_V	AR_V1	_NOBS_	_TAU_	_TREND_	_DLAG_	_PVALUE_
1	PARM	0 Converged	AR_V		0.24543	0.28522	-1	-0.056220	0.31506	79	-1.97643	1	1	0.29642
2	COV	0 Converged	AR_V	Intercept	0.24543	0.03171	.	-0.004802	0.00198	79	-1.97643	1	1	0.29642
3	COV	0 Converged	AR_V	DLAG_V	0.24543	-0.00480	.	0.000809	-0.00015	79	-1.97643	1	1	0.29642
4	COV	0 Converged	AR_V	AR_V1	0.24543	0.00198	.	-0.000152	0.01123	79	-1.97643	1	1	0.29642

Dickey-Fuller test for 3 year bond rate

Obs	f	df	df1	ddf	b	db	db1	ddb
1	8.47667	.	.	.	10.6767	.	.	.
2	7.92333	-0.55334	.	.	9.7633	-0.91337	.	.
3	7.90000	-0.02333	-0.55334	0.53001	9.2867	-0.47666	-0.91337	0.43671
4	8.10333	0.20333	-0.02333	0.22666	8.8433	-0.44334	-0.47666	0.03332
5	7.82667	-0.27666	0.20333	-0.47999	7.9367	-0.90666	-0.44334	-0.46332
6	6.92000	-0.90667	-0.27666	-0.63001	7.1800	-0.75667	-0.90666	0.14999
7	6.20667	-0.71333	-0.90667	0.19334	6.6567	-0.52333	-0.75667	0.23334
8	6.26667	0.06000	-0.71333	0.77333	6.4833	-0.17334	-0.52333	0.34999
9	6.22000	-0.04667	0.06000	-0.10667	6.5167	0.03334	-0.17334	0.20668
10	6.65000	0.43000	-0.04667	0.47667	7.7200	1.20333	0.03334	1.16999

Dickey-Fuller test for differenced federal funds series

The AUTOREG Procedure

Dependent Variable	ddf

Dickey-Fuller test for differenced federal funds series

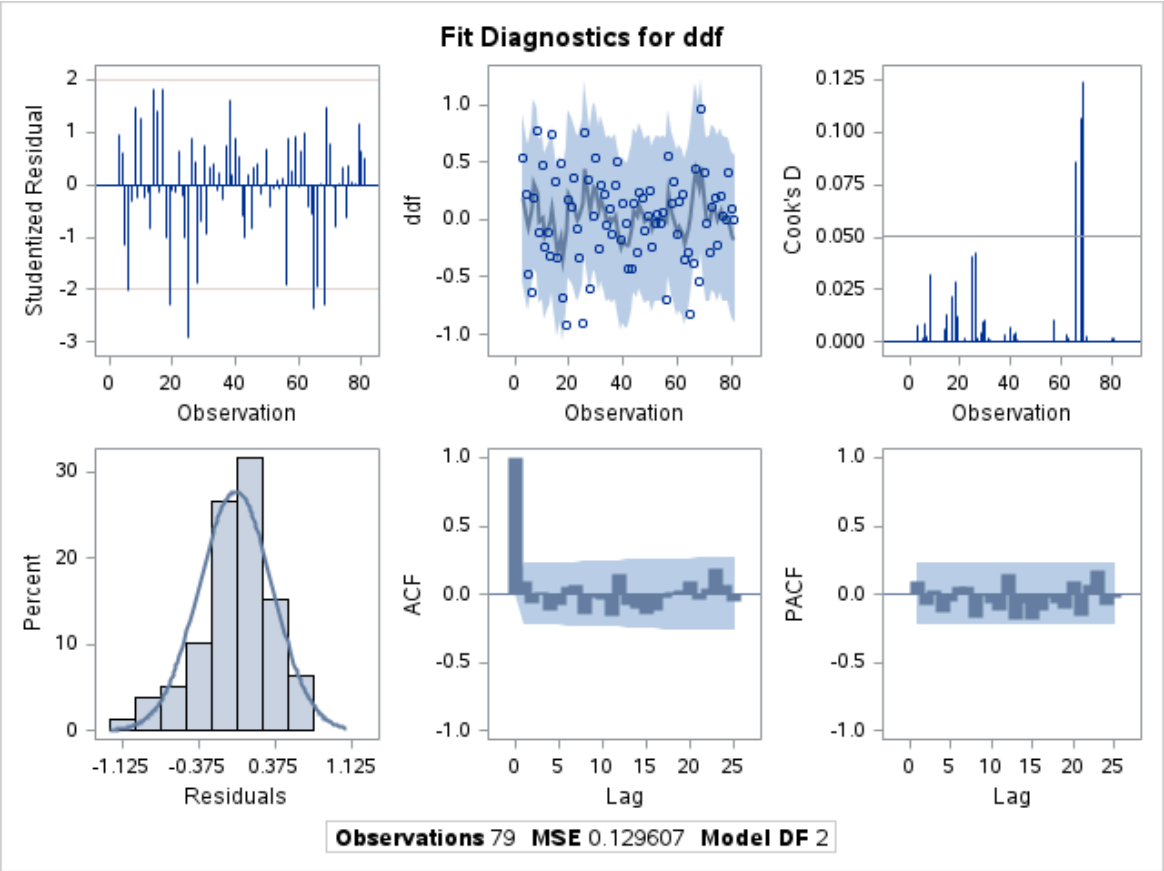
The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	10.1093835	DFE	78
MSE	0.12961	Root MSE	0.36001
SBC	66.1390157	AIC	63.7695679
MAE	0.26803637	AICC	63.8215159
MAPE	575.378341	HQC	64.7188412
Durbin-Watson	1.7988	Total R-Square	0.1707
NOTE: No intercept term is used. R-squares are redefined.			

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
df1	1	-0.3405	0.0850	-4.01	0.0001

Dickey-Fuller test for differenced federal funds series

The AUTOREG Procedure



Dickey-Fuller test for differenced fed funds series

Obs	_TYPE_	_STATUS_	_DEPVAR_	_NAME_	_MSE_	Intercept	AR_V	DLAG_V	_NOBS_	_TAU_	_TREND_	_DLAG_	_PVALUE_
1	PARM	0 Converged	AR_V		0.12961	.	-1	-0.34047	79	-4.00735	0	1	.000109725

Obs	_TYPE_	_STATUS_	_DEPVAR_	_NAME_	_MSE_	Intercept	AR_V	DLAG_V	_NOBS_	_TAU_	_TREND_	_DLAG_	_PVALUE_
2	COV	0 Converged	AR_V	DLAG_V	0.12961	.	.	0.00722	79	-4.00735	0	1	.000109725

Dickey-Fuller test for differenced 3 year bond rate series

The AUTOREG Procedure

Dependent Variable	ddb

Dickey-Fuller test for differenced 3 year bond rate series

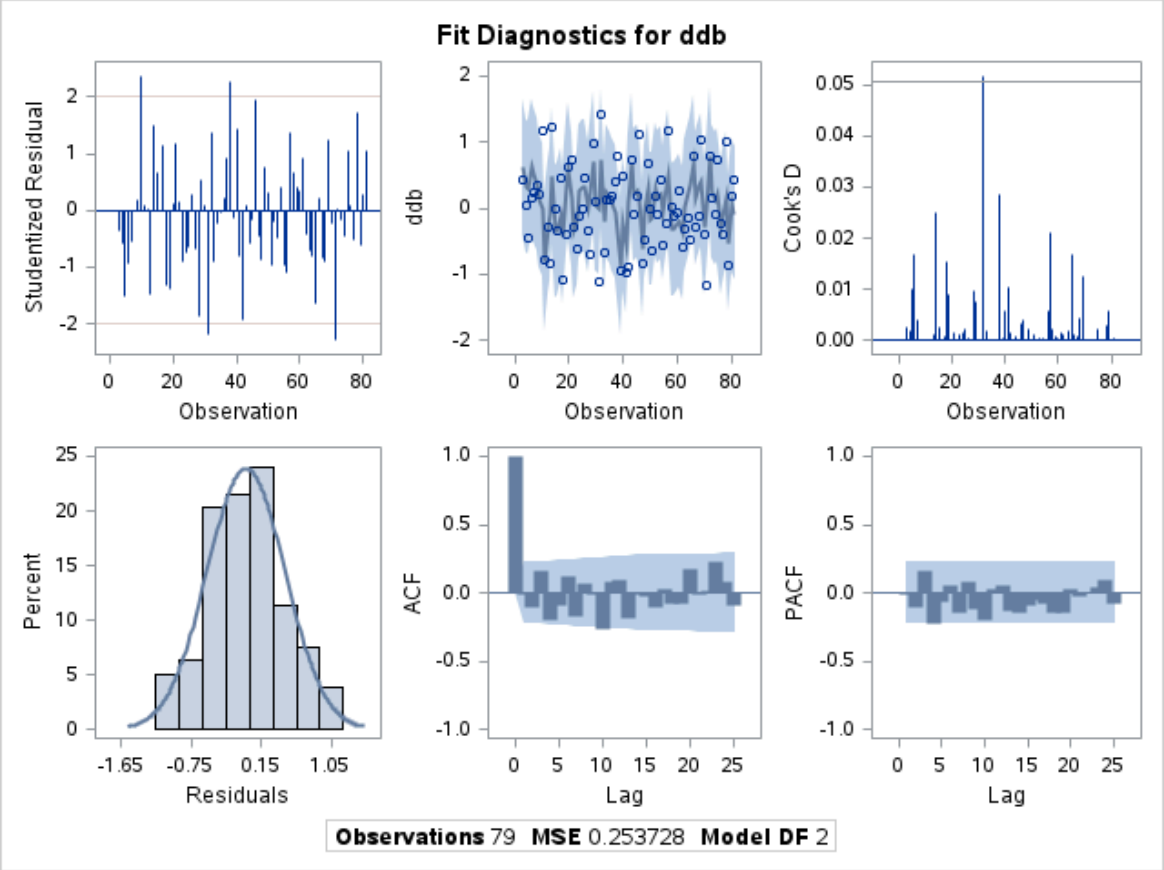
The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	19.7907697	DFE	78
MSE	0.25373	Root MSE	0.50371
SBC	119.207392	AIC	116.837945
MAE	0.39435231	AICC	116.889893
MAPE	164.550119	HQC	117.787218
Durbin-Watson	1.9804	Total R-Square	0.3454
NOTE: No intercept term is used. R-squares are redefined.			

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
db1	1	-0.6791	0.1059	-6.42	<.0001

Dickey-Fuller test for differenced 3 year bond rate series

The AUTOREG Procedure



Dickey-Fuller test for differenced 3 year bond rate series

Obs	_TYPE_	_STATUS_	_DEPVAR_	_NAME_	_MSE_	Intercept	AR_V	DLAG_V	_NOBS_	_TAU_	_TREND_	_DLAG_	_PVALUE_
1	PARM	0 Converged	AR_V		0.25373	.	-1	-0.67914	79	-6.41504	0	1	.000004128
2	COV	0 Converged	AR_V	DLAG_V	0.25373	.	.	0.01121	79	-6.41504	0	1	.000004128

save residuals to test for cointegration

The AUTOREG Procedure

Dependent Variable	b

save residuals to test for cointegration

The AUTOREG Procedure

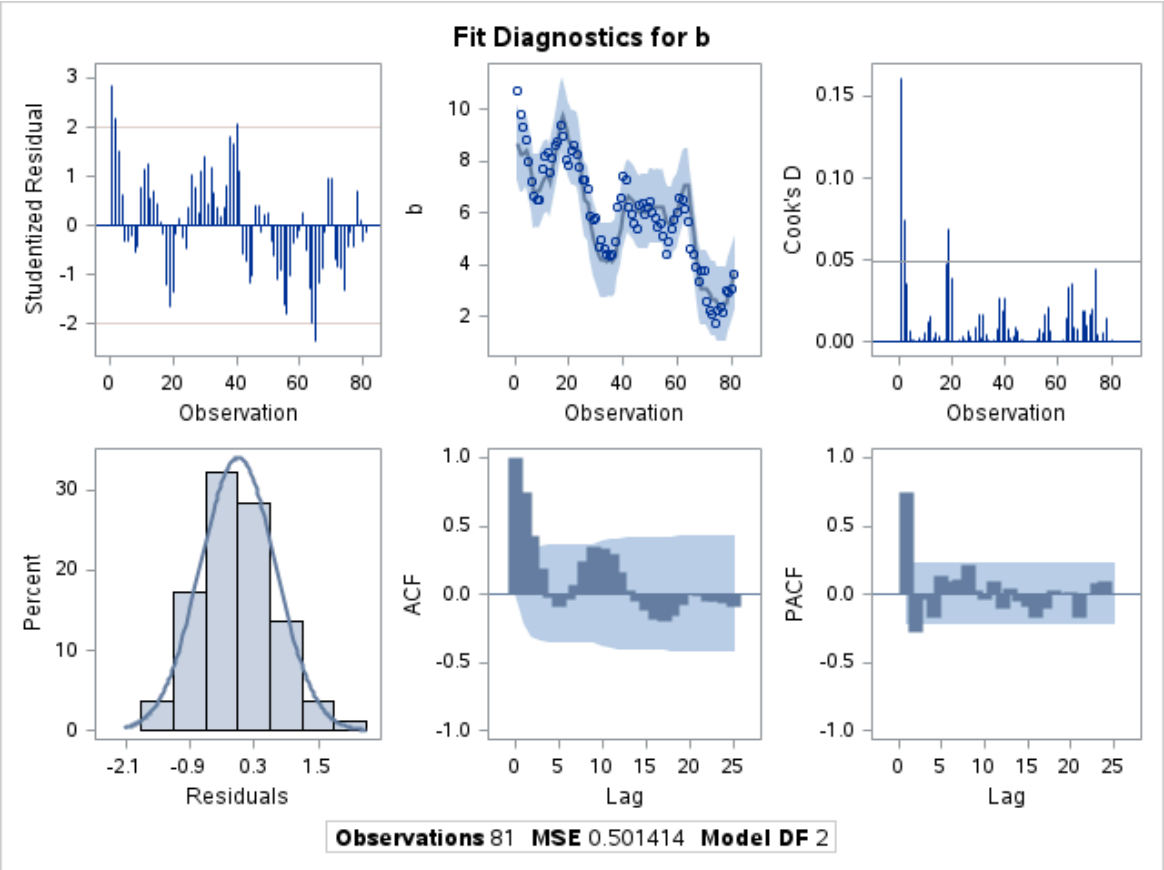
Ordinary Least Squares Estimates			
SSE	39.6116996	DFE	79
MSE	0.50141	Root MSE	0.70811
SBC	180.715645	AIC	175.926747
MAE	0.56028338	AICC	176.080593
MAPE	10.5747896	HQC	177.848116
Durbin-Watson	0.4139	Total R-Square	0.8807

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
Intercept	1	1.6437	0.1948	8.44	<.0001

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
f	1	0.8325	0.0345	24.15	<.0001

save residuals to test for cointegration

The AUTOREG Procedure



test for cointegration

The AUTOREG Procedure

Dependent Variable	dehat

test for cointegration

The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	12.2777309	DFE	77
MSE	0.15945	Root MSE	0.39931
SBC	85.8599867	AIC	81.121091
MAE	0.31543164	AICC	81.2789857
MAPE	223.103917	HQC	83.0196376
Durbin-Watson	2.0022	Total R-Square	0.2416
NOTE: No intercept term is used. R-squares are redefined.			

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
ehat1	1	-0.3143	0.0692	-4.54	<.0001
dehat1	1	0.3147	0.1022	3.08	0.0029

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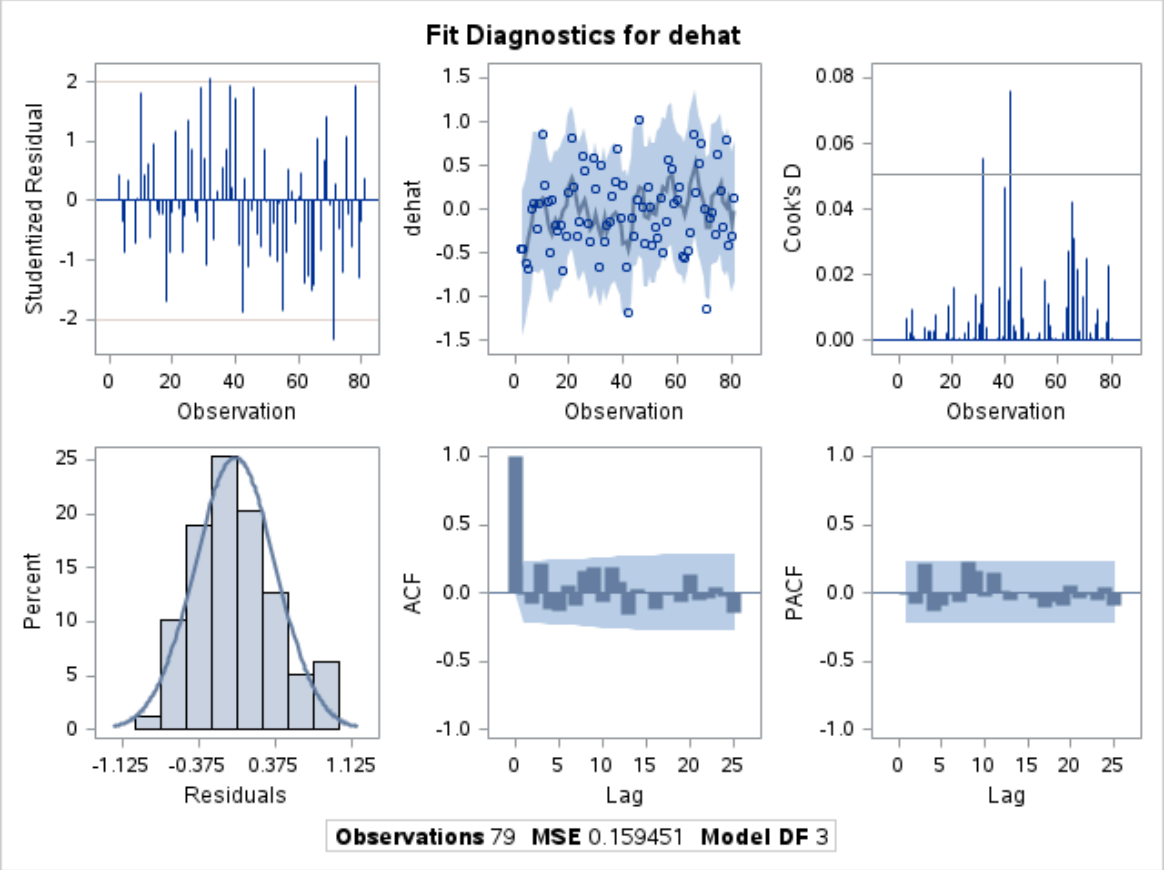
test for cointegration

The AUTOREG Procedure